NEOSTATS · AI ENGINEER ASSIGNMENT

# AI-Powered Credit Risk Intelligence Platform

An end-to-end, explainable credit-risk platform on the Home Credit Default Risk dataset: exploratory analysis, a calibrated risk model, SHAP explanations, machine-derived credit policy, and a guarded natural-language interface to the data.

| 307,511 | 0.782 | 15× | 0 |
|---|---|---|---|
| APPLICANTS MODELLED | ROC-AUC | MORE DEFAULTS CAUGHT | API KEYS REQUIRED |

THE PROBLEM

# A wrong approval costs far more than a wrong decline

- **Only 8.07% of applicants default.** Accuracy is meaningless — predicting "everyone repays" scores 91.9%.
- **The product is a probability, not a ranking.** "0.20" must mean one in five actually default, or pricing and provisioning are built on sand.
- **Every decision must be explainable** — to the applicant who was declined, and to a regulator months later.
- **Analysts need the data in plain English**, without waiting on an engineer to write SQL.

### What "good" looks like

A **calibrated** probability, a decision threshold **tuned** to the cost of being wrong, a reason the applicant can act on, and a policy a credit committee can audit — all runnable by one person with one command.

### Why the imbalance drives everything

An 11.4:1 class ratio decides the metric (PR-AUC, not accuracy), the imbalance strategy (weighting, not SMOTE) and the threshold (tuned, never 0.5). Those three choices shape the entire pipeline.

WHAT WAS BUILT

# Six capabilities, one application

### 1 · Exploratory analysis

Dataset summary, feature categorisation by type and business domain, data-quality findings, and **9 business insights** — each a chart plus an actionable takeaway with 95% confidence intervals.

### 2 · Risk scoring

Three-model bake-off under identical CV, calibrated probability, 0–1000 risk score, Low/Medium/High band and a recommended action.

### 3 · Explainability

SHAP per applicant, rendered as a **plain-English narrative** plus a signed contribution chart, and a portfolio-level driver ranking.

### 4 · Credit policy

A shallow surrogate tree fitted to the model's own predictions, exported as readable IF/THEN rules — with fidelity reported.

### 5 · Talk to your data

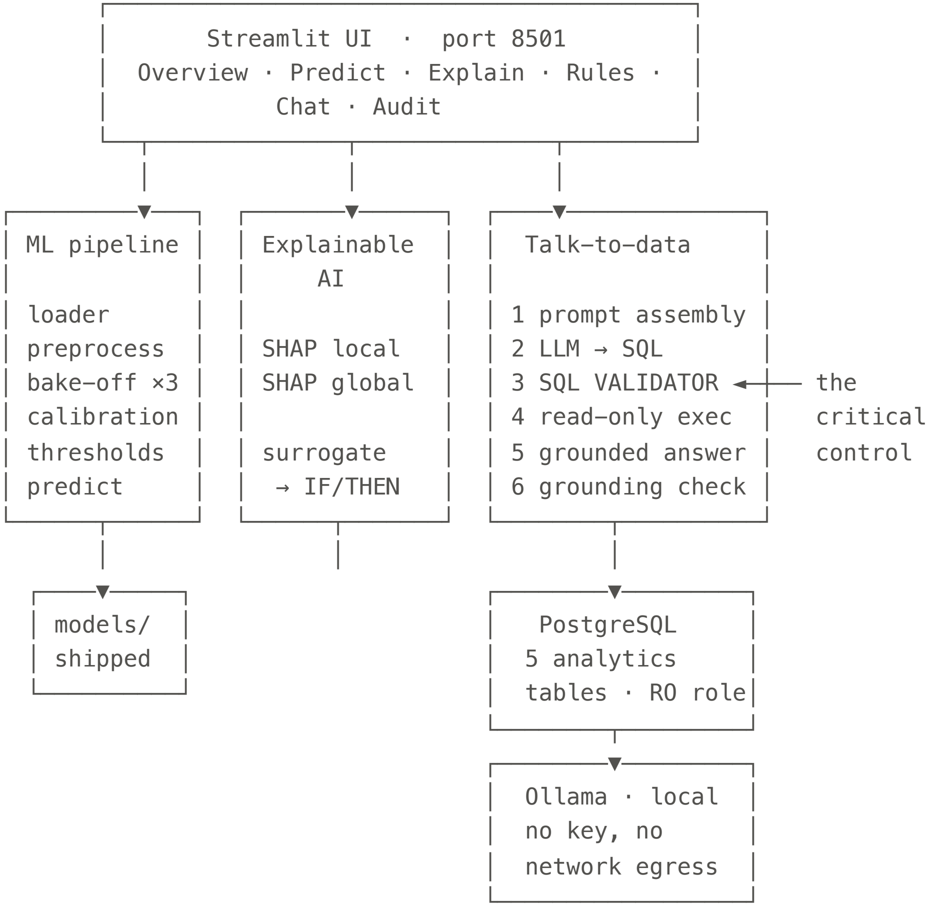
Plain-English questions become validated, read-only SQL. Three independent safety layers plus numeric grounding of every answer.

### 6 · Audit trail

Every credit decision and every question recorded append-only — including the **reasons given**, so a decision is reconstructable later.

ARCHITECTURE

# Three containers, one command




### Data flow

CSV → join + feature engineering → model artifacts → analytics tables → UI and chat. The chat never touches the model artifacts; it queries the database, which includes the model's own predictions as a table.

### Deployment

`docker compose up` brings up all three. A second profile, `docker-compose.lite.yml`, drops the local model runtime: **1.83 GB against 10.43 GB**, and 267 MB RAM instead of ~6 GB.

DATA

# Four of seven tables — chosen by measurement

Each auxiliary block was switched on in turn and its contribution measured, rather than assumed from what Kaggle solutions use. LightGBM, 3-fold, out-of-fold, on all 307,511 rows.

| CONFIGURATION | FEATURES | PR-AUC | ROC-AUC | MARGINAL GAIN |
|---|---|---|---|---|
| application only | 139 | 0.2507 | 0.7660 | — |
| + bureau — credit held elsewhere | 166 | 0.2576 | 0.7697 | **+0.0069** |
| + previous_application — our own prior decisions | 183 | 0.2638 | 0.7747 | **+0.0062** |
| **+ installments_payments — repayment behaviour** | **198** | **0.2708** | **0.7800** | **+0.0070** |

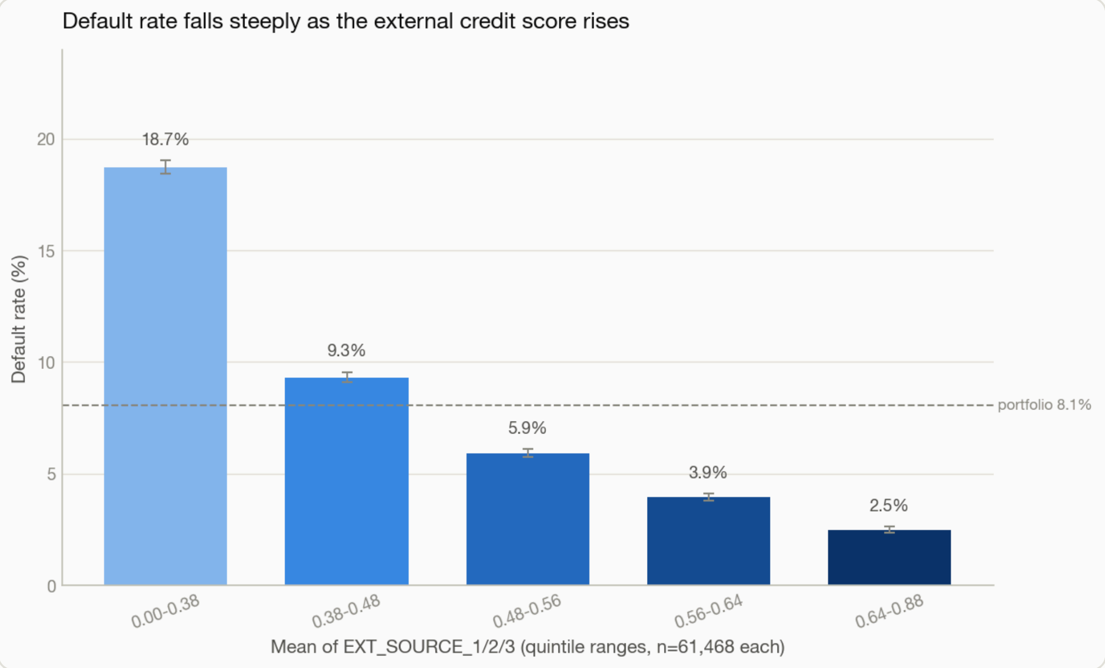
### Why each stays

All three contribute roughly equally (+0.007 PR-AUC each, +0.020 total). The three monthly panels — bureau_balance, POS_CASH, credit_card — are excluded as largely redundant with these aggregates. Nothing is included merely because it exists.

### The one that changed the analysis

**installments_payments** is the table that actually contains *repayment behaviour* — payment date against due date, amount paid against amount owed. The brief names it as an analysis area, and no other table substitutes for it.

# External credit scores dominate — and monotonically

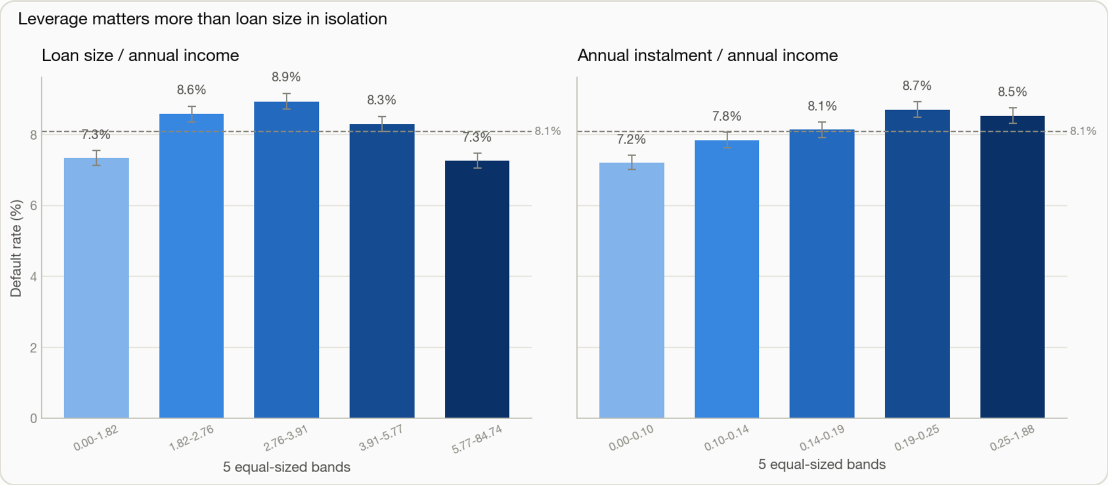



Error bars are 95% Wilson intervals. Every rate in the analysis carries one, and each written takeaway hedges automatically when intervals overlap.

- Default rate falls **18.7%** → **2.5%** across evenly sized quintiles — a 7.6× spread.
- The decline is **monotonic across every band**, which is what makes it trustworthy for policy rather than merely correlated.
- EXT_SOURCE_1 is missing for the majority of applicants — hence the model uses the **mean, min and count** of the three scores rather than any single one.

**Acted on:** the bottom band warrants manual review or a pricing uplift, and an applicant missing all three external scores should be routed to manual assessment rather than scored on the remaining fields.

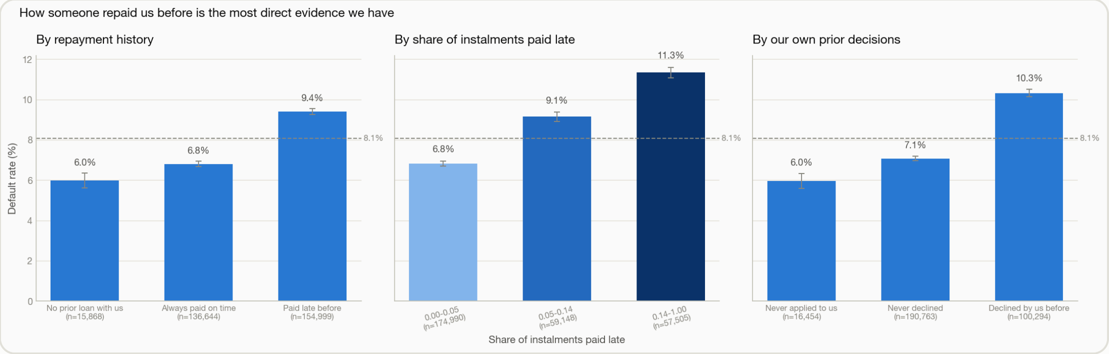
# The finding that contradicts the obvious assumption




- **Loan-to-income is an inverted U** — 7.3% at *both* extremes, peaking at 8.9% in the middle. The most heavily leveraged applicants are **not** the riskiest.
- The likely reason: very high ratios pick up secured and longer-term products rather than distressed borrowing.
- **Instalment-to-income does rise** consistently, 7.2% → 8.7%. Only the second is a usable affordability constraint.

> **Why this matters methodologically:** an earlier version of this analysis asserted the opposite, because the takeaway was written by hand. It now derives its claim from the data by rank correlation — and reaches opposite, correct conclusions on the real and synthetic datasets.

# The most *defensible* signal is not the largest one




- Applicants who ever paid one of **our** instalments late default at **9.4%**, against 6.8% for those who always paid on time.
- Applicants with arrears on **external** credit default at **15.9%** against 7.6% — 2.1×, non-overlapping intervals.
- “No prior loan with us” is kept as its own category, not folded into “paid on time”. Absence of history is a distinct risk state.

> Every other strong feature is a **proxy** — education stands in for income stability, an external score summarises another institution’s judgement. This is the applicant’s own conduct on their own obligations, recorded by us and auditable. It survives fair-lending scrutiny in a way demographic proxies do not.

# Three candidates, identical folds, a statistical tie-break

| MODEL | PR-AUC | ± S.E. | ROC-AUC | BRIER | FIT | TREE SHAP |
|---|---|---|---|---|---|---|
| **LightGBM — selected** | **0.2720** | **0.0033** | **0.7811** | **0.1711** | **33s** | **Yes** |
| CatBoost | 0.2705 | 0.0039 | 0.7809 | 0.1801 | 209s | Yes |
| Logistic regression | 0.2480 | 0.0026 | 0.7666 | 0.1961 | 36s | No |

### Why LightGBM

LightGBM and CatBoost are **statistically tied** — a 0.0015 gap against standard errors of 0.0033 and 0.0039. Any candidate within one standard error of the leader is treated as tied, and the tie goes to exact tree-SHAP support, then fit cost. Same performance in **a sixth of the time**.

### Class imbalance: weighting, not SMOTE

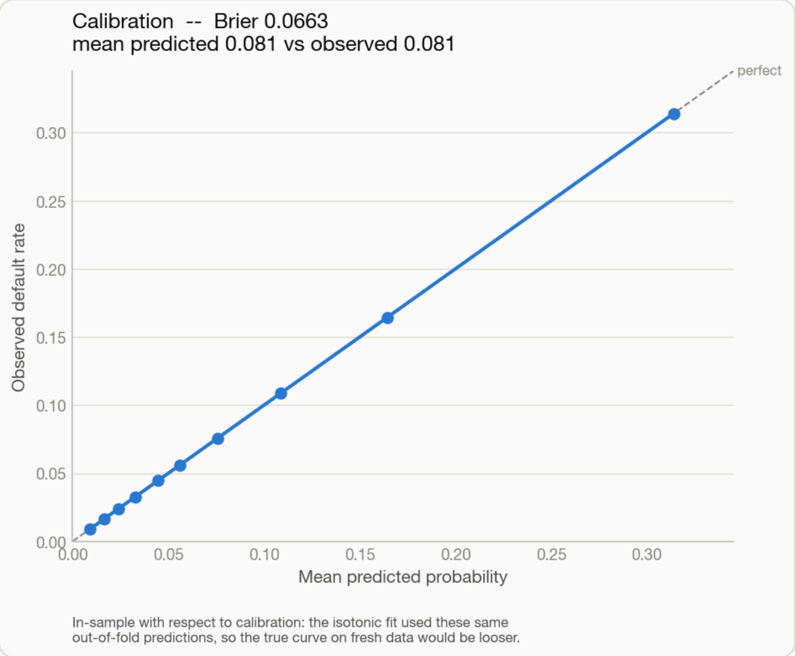
SMOTE interpolates synthetic minority rows, which shifts the class prior and biases probabilities upward. The product here *is* a calibrated probability, so that is disqualifying regardless of what it does to AUC.

### Two bugs found before the result was trusted

**1.** LightGBM silently stopped at **iteration 5**. It tracks its objective's default metric alongside the requested one and halts when *any* stalls; under `scale_pos_weight` the unweighted logloss degrades immediately. Worth **0.07 PR-AUC**.

**2.** On synthetic fixtures logistic regression won — because the generating process was linear in the logit, making it correctly specified *by construction*. On real data the boosters win by seven standard errors. The selection logic never changed; the data did.

# A score that means something, at a threshold that is not 0.5



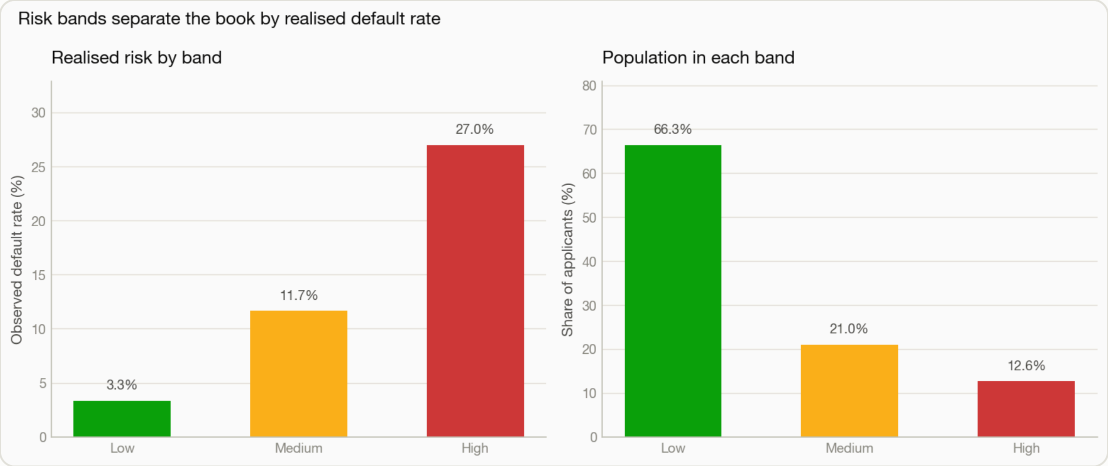

**Brier 0.1711** → **0.0663** after isotonic calibration, with mean predicted matching observed exactly. Stated on the chart itself: the calibrator was fitted on these same out-of-fold predictions, so the curve is in-sample with respect to calibration.

**At an 8% base rate, a 0.5 cut-off approves almost everyone.** Both the decision point and the band edges are tuned on out-of-fold predictions.

| | TUNED (0.165) | NAIVE (0.5) |
|---|---|---|
| Precision | 0.267 | 0.604 |
| Recall | **0.430** | 0.029 |
| Share of book flagged | 13.0% | 0.4% |
| **Defaults caught** | **10,665** | **713** |
| Defaults missed | 14,160 | 24,112 |

> The naive cut-off looks more *precise* only because it flags almost nobody — it finds 3% of the defaults in the book. That is not a usable credit policy.

# The top eighth of the book holds two-fifths of the defaults

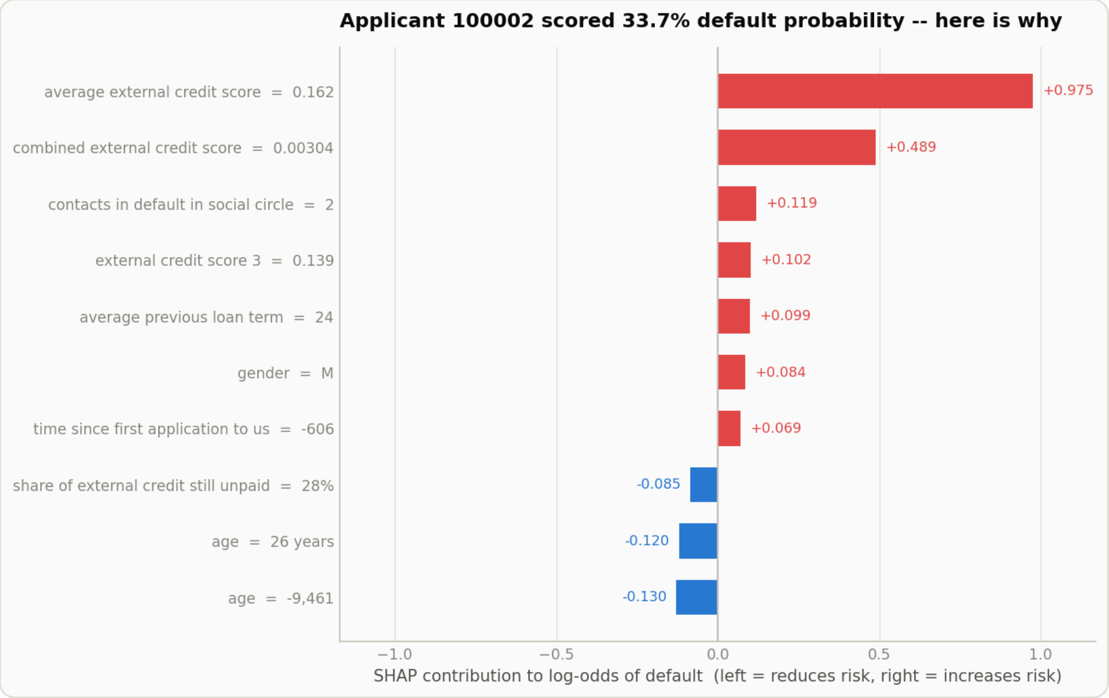



| BAND | POPULATION | DEFAULT RATE | SHARE OF DEFAULTS |
|---|---|---|---|
| Low | 66.3% | **3.34%** | 27% |
| Medium | 21.0% | 11.65% | 30% |
| **High** | **12.6%** | **26.96%** | **42%** |

### Bands bound the *marginal* applicant

Low is defined as “at or below the portfolio base rate”. An earlier version set the edge wherever the *average* rate below it met a 5% target — which let a wide band hide its own upper end and put a rule predicting 12.9% default into the band labelled “Low”. Reading the generated policy rules exposed it.

# An explanation the applicant could actually read




Blue reduces risk, red increases it. Plain-English labels throughout — never raw column names.

> "This applicant has a **87.5%** estimated probability of default, placing them in the **High** risk band. Recommended action: refer for review. The main factors increasing risk are that their **average external credit score of 0.171** raises the risk, their combined external credit score of 3.25e-05 raises the risk, and their **share of instalments paid late of 48%** raises the risk."

- A ranked table of log-odds is an explanation for a modeller. Someone who has been **declined** is owed something they can act on.
- SHAP and the policy rules draw from **one shared vocabulary**, so the two can never describe the same feature differently.
- **No prediction claims certainty.** Isotonic calibration returns exactly 0 and 1 on pure bins, which produced "100.0% probability of default" — indefensible from a finite sample. Probabilities are bounded to [0.001, 0.999].

BUSINESS-READABLE DECISION RULES

# The policy the model is implicitly applying

```
RULE 1 — covers 1,930 applicants (3.9% of the book)

  IF   average external credit score <= 0.395
  AND  combined external credit score <= 0.023
  AND  average external credit score <= 0.247

  THEN predicted default risk 31.2%
       → High risk band

       observed default rate
       in this group: 30.7%
```

13 rules from a depth-4 surrogate tree, fitted to 50,000 scored applicants. Full set in `reports/policy_rules.csv`.

- The tree is fitted to the model's **own calibrated predictions** — not to the labels. The goal is to describe what the model does, *including where it is wrong*.
- Predicted and observed agree to within about a percentage point on every leaf. That agreement is the evidence a rule means what it claims.
- Categorical splits render as membership tests — "employment type is Pensioner" — rather than thresholds on an integer code.

### Fidelity is reported with every rule set

**$R^2$ 0.556 · band agreement 73.1%.** Roughly one applicant in four would be banded differently by the rules than by the model. Stated plainly, because a surrogate that does not track the model is worse than no surrogate — it looks authoritative while being wrong. The rules describe the policy; they do not replace the model.

# An LLM writes the SQL, so the SQL is untrusted input

### Layer 1 · SQL validator

Parses with sqlglot. Single `SELECT` only; every write/DDL node rejected **anywhere** in the tree — including a `DELETE` hidden inside a CTE; table and column whitelist checked against the **live** schema; row cap injected. The statement that runs is **re-serialised from the parse tree**, never the model's string.

### Layer 2 · Read-only role

A separate PostgreSQL role with `SELECT` and nothing else, created at first start with `ALTER DEFAULT PRIVILEGES`. Verified live: `SELECT` returns 307,511 rows; `DELETE` and `DROP` are refused by the database itself. A statement timeout bounds anything valid but pathological.

### Layer 3 · Numeric grounding

Every figure quoted in an answer must occur in the returned rows. The most dangerous failure is not bad SQL — it is a **fluent summary containing an invented number**, which looks exactly like a correct answer and which no SQL validator can catch.

### The failure this control exists for

`“…approximately 0.08% of the total sample size (2,854 + 50 + 153 + 942 = ` **`3,999`** `).”`

Every input real; the arithmetic and the conclusion invented. Both fabricated figures are caught, the summary is discarded, and a truthful description is shown instead.

### Layer 0 · Refusal

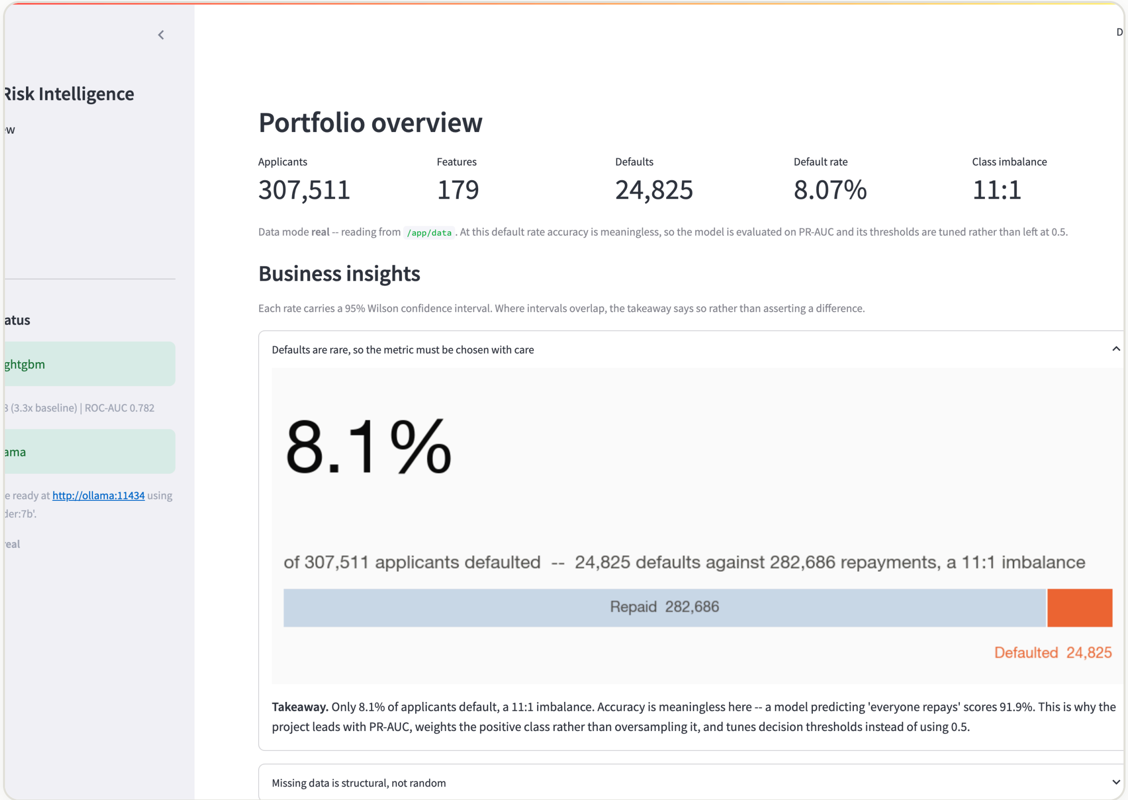
**“What is the average credit card balance?”**

→ *“I can’t answer that from the available data. There is no column in the schema that represents a credit card balance.”*

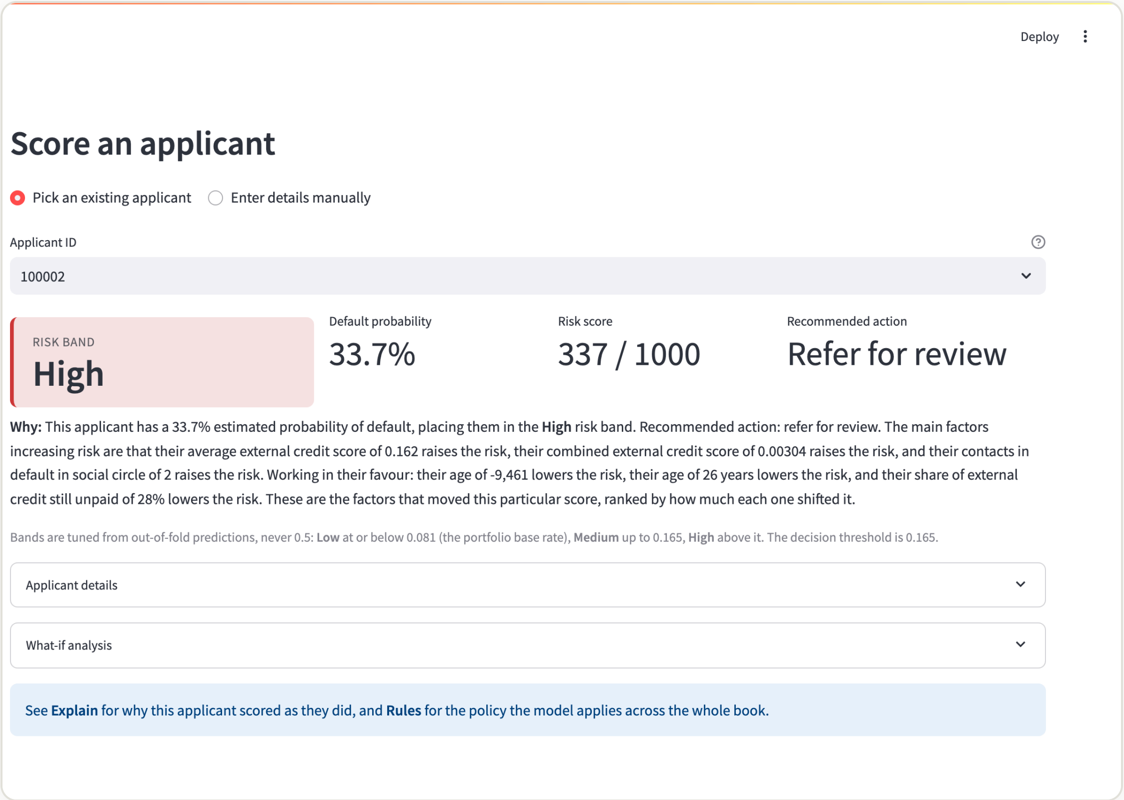
An admitted gap beats an invented column. The refusal is believed, not retried.

THE APPLICATION

# End-to-end in one interface

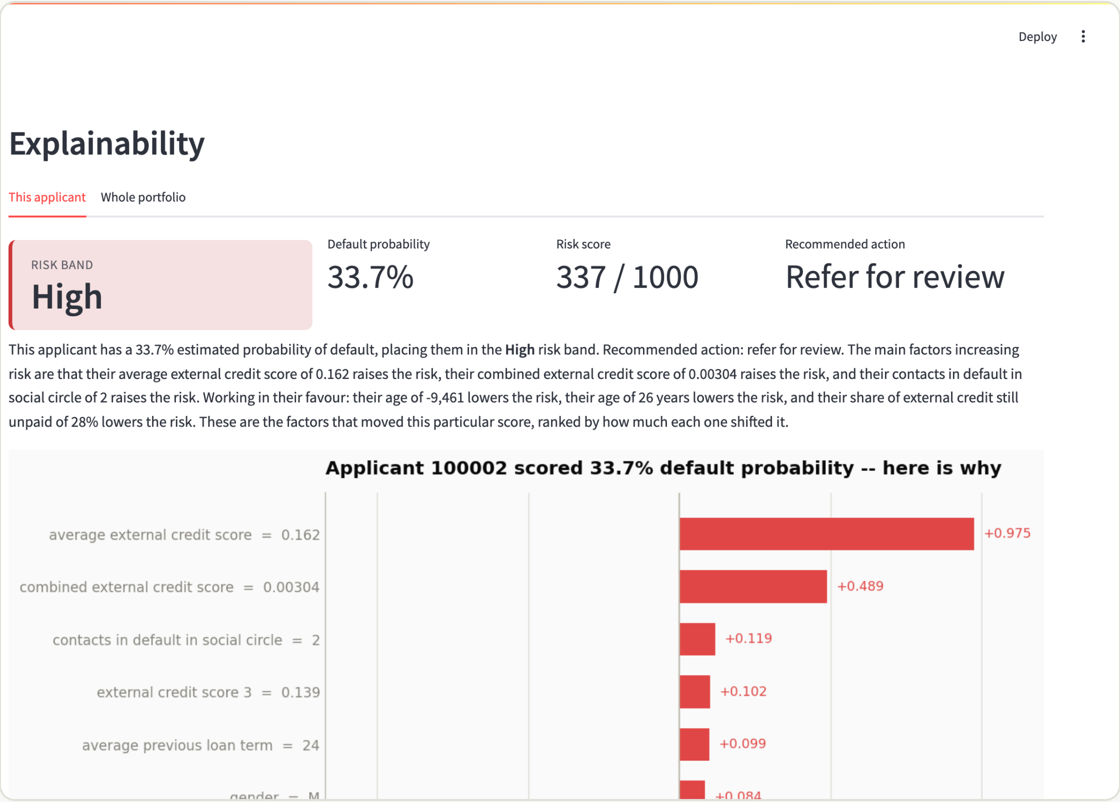



**Overview** — portfolio summary and the 9 insights

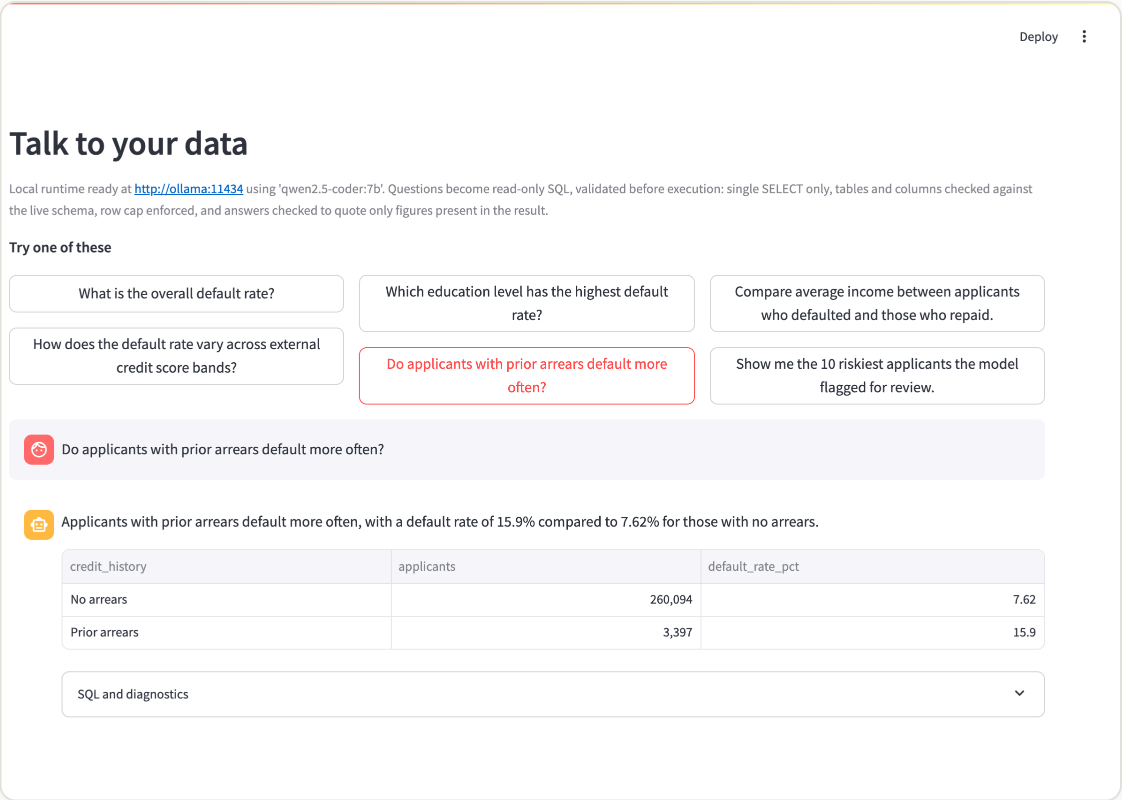



**Predict** — band, score, action, plus what-if analysis

# Explanations, policy and plain-English querying




**Explain** — narrative, SHAP chart and ranked reasons




**Chat** — question, grounded answer, rows and the SQL behind it

# On the full 307,511-row dataset

| 0.782 | 0.268 | 0.066 | 10,665 |
|---|---|---|---|
| ROC-AUC | PR-AUC · 3.3× BASELINE | BRIER, CALIBRATED | DEFAULTS CAUGHT VS 713 |

### Model

LightGBM, selected over CatBoost on a one-standard-error tie-break resolved by explainability and fit cost. 198 engineered features from 4 tables. Hyperparameters from a documented randomised search.

### Engineering

**300 tests**, all runnable with no database, no model runtime and no API key. CI on every push. Two deployment profiles. A fresh clone scores applicants in under 20 seconds.

### Safety

33 injection and hallucination attacks attempted, **0 leaks**. Read-only role verified against a live database. Every decision and query recorded to an append-only audit log.

HONEST ASSESSMENT

# Limitations, and what I would do next

### Known limitations

- **Calibration is measured in-sample.** The isotonic fit uses the same out-of-fold predictions the reliability curve is drawn from, so that curve is optimistic. A held-out set would measure it honestly.
- **The surrogate is an approximation.** $R^2$ 0.556 means roughly one applicant in four is banded differently by the rules than by the model.
- **Three tables excluded on structural judgement**, not measured the way the other three were.
- **No fairness audit.** Gender and family status are legally sensitive in lending; disparate-impact testing is required before any production use.

### Next, in value order

- A held-out calibration split, and calibration drift monitoring.
- Fairness metrics per protected attribute, with reject-option post-processing.
- A labelled question→SQL evaluation set scored on every prompt change, turning prompt edits into a measurable experiment rather than a judgement call.
- Model monitoring — PSI on feature distributions, alerting on score drift.
- Authentication and per-user query audit for the chat.

> Naming real limitations is part of the deliverable. A credit model whose weaknesses are undocumented is not safer — only less examined.